Project Baseline

Metrics

Metric	Value	Rating
Performance	45	🔴 Poor
Accessibility	91	🟢 Good
Best Practices	77	🟡 Needs Improvement
SEO	100	🟢 Good
First Contentful Paint (FCP)	0.7 s	🟢 Good
Largest Contentful Paint (LCP)	2.6 s	🟡 Needs Improvement
Speed Index	12.9 s	🔴 Poor
Total Blocking Time (TBT)	6,660 ms	🔴 Poor

Cumulative Layout Shift (CLS)	0.004	🟢 Good

The Lighthouse audit shows that rendering performance is the weakest area of the website. While Accessibility and SEO perform well, the low Performance score indicates significant opportunities to improve loading speed and responsiveness.

Networking

Initial (Hard) Reload

Metric	Value
Requests	**771**
Data Transferred	**8.2 MB**
Total Resource Size	**72.2 MB**
Load Time	**6.3 s**

Soft Reload

Metric	Value
Requests	**628**
Data Transferred	**635 kB**
Total Resource Size	**92.3 MB**
Load Time	**5.4 s**

Cache Savings

- Initial transfer: **8.2 MB**
- Repeat transfer: **635 kB**

- **≈92% reduction in transferred data**

This demonstrates that browser caching is effective for returning visitors.

Resource Breakdown

Resource	Size	Approx. Share
JavaScript	~25.0 MB	~35%
CSS	~2.2 MB	~3%
Images	~1.0 MB	~1.5%

JavaScript represents by far the largest resource category and contributes significantly to download, parsing, and execution time. JavaScript assets are compressed with gzip, reducing network transfer costs.

Corrective Findings

Finding 1 — Excessive JavaScript Payload and Main-Thread Work

Observation

The homepage loads approximately **25 MB of JavaScript**, making it the largest resource category. Lighthouse also reports a **Total Blocking Time of 6,660 ms**, indicating extensive JavaScript execution before the page becomes fully interactive.

User Impact

Users may see page content but experience delays when scrolling, clicking buttons, opening menus, or interacting with the website.

Metrics Affected

- Total Blocking Time (TBT)
- Performance Score
- Largest Contentful Paint (LCP)
- Interaction responsiveness

Most Likely Cause

Large JavaScript bundles and significant client-side processing during the initial page load.

Recommended Solution

- Remove unused JavaScript.
- Split large bundles into smaller chunks.
- Defer non-essential scripts.
- Lazy-load functionality that is not required immediately.

Finding 2 — Images Are Larger Than Necessary

Observation

The audit estimates **1,596 KiB** of potential image savings. Several images are downloaded at much higher resolutions than required.

User Impact

Large images increase loading time, particularly on slower networks and mobile devices.

Metrics Affected

- Largest Contentful Paint (LCP)
- Speed Index
- Performance Score

Most Likely Cause

Oversized images and insufficient image compression.

Recommended Solution

- Compress images further.
- Use WebP or AVIF.
- Serve appropriately sized images for each device.

Finding 3 — Excessive Number of Network Requests

Observation

The homepage generates **771 requests** during the initial load.

User Impact

Each request introduces additional network overhead and latency, particularly for mobile users.

Metrics Affected

- Overall page load time
- Speed Index
- LCP

Most Likely Cause

Many JavaScript files, third-party services, API calls, fonts, and other assets.

Recommended Solution

- Remove unnecessary assets.
- Combine small resources where appropriate.
- Delay non-critical requests until after the initial page load.

Finding 4 — Initial Download Size Is High

Observation

The initial page load transfers **8.2 MB** of data.

User Impact

Large downloads increase loading time and consume more mobile data.

Metrics Affected

- Largest Contentful Paint (LCP)

- Speed Index
- Performance Score

Most Likely Cause

Large JavaScript bundles, media assets, and third-party resources.

Recommended Solution

Reduce JavaScript size, optimize media assets, and remove unnecessary resources.

Finding 5 — Browser Cache Lifetime Can Be Improved

Observation

Although caching works well, Lighthouse estimates an additional **1,128 KiB** could be saved by improving cache lifetimes.

User Impact

Returning users still re-download assets that could remain cached longer.

Metrics Affected

- Largest Contentful Paint (LCP)
- Speed Index

Most Likely Cause

Short cache lifetimes for some static assets.

Recommended Solution

Increase cache durations for static images, CSS, JavaScript, and fonts where appropriate.

Finding 6 – Overall Rendering Performance Remains Poor

Observation

The Lighthouse audit reports an overall **Performance score of 45**, which falls within the **Poor** category. While the website performs well in Accessibility (91) and SEO (100), rendering performance remains significantly below recommended levels, indicating that users may experience slow loading and reduced responsiveness.

User Impact

Users may experience slower page loads and delayed responsiveness, particularly on slower internet connections or less powerful devices. This can negatively affect the overall browsing experience and increase the likelihood of users leaving the site before it becomes fully usable.

Metrics Affected

- Performance Score
- Largest Contentful Paint (LCP)
- Speed Index
- Total Blocking Time (TBT)

Most Likely Cause

The low performance score is primarily the result of several combined issues, including heavy JavaScript execution, oversized images, a high number of network requests, and a large initial download size.

Recommended Solution

Prioritize the improvements identified in this report by reducing JavaScript execution, optimizing image delivery, minimizing unnecessary network requests, and reducing the overall amount of data transferred during the initial page load.

Good Findings

Good Finding 1 — Effective Browser Caching

A soft refresh transfers only **635 kB** compared with **8.2 MB** during the initial load, representing approximately **92% less transferred data**. This demonstrates that browser caching is functioning effectively and provides much faster repeat visits.

Good Finding 2 — HTTP Compression Is Enabled

Response headers indicate that JavaScript assets are compressed using **gzip**, significantly reducing the amount of data transferred over the network. This improves loading performance, especially for users on slower connections.

Overall Conclusion

The Wayfair homepage demonstrates strong Accessibility and SEO scores, along with effective browser caching and HTTP compression. However, overall performance is limited by heavy JavaScript execution, large initial network transfers, oversized images, and a very high number of network requests. Addressing these issues would improve loading speed, responsiveness, and the overall user experience, particularly for mobile users and those on slower network connections.